Statistics & Probability

ANSWER KEY

Mean, median, mode, and range

1 Find the mean, median, and range of the data set: $12, 18, 15, 22, 9, 18$.

Mean $= \dfrac{12+18+15+22+9+18}{6} = \dfrac{94}{6} \approx 15.67$. Sorted: $9, 12, 15, 18, 18, 22$. Median (average of the two middle values): $\dfrac{15+18}{2} = 16.5$. Range: $22 - 9 = 13$.

2 A data set of 5 values has a mean of 14. If four of the values are $10, 12, 16, 19$, find the fifth value.

Sum of all 5 values: $14 \times 5 = 70$. Sum of known values: $10 + 12 + 16 + 19 = 57$. Fifth value: $70 - 57 = 13$.

Weighted averages

3 A class of 20 students has an average test score of 78, and another class of 30 students has an average of 85. Find the combined average score of all 50 students.

Total points from class 1: $20 \times 78 = 1560$. Total points from class 2: $30 \times 85 = 2550$. Combined total: $1560 + 2550 = 4110$. Combined average: $\dfrac{4110}{50} = 82.2$.

Basic probability

4 A bag contains 6 red, 4 blue, and 5 green marbles. Find the probability of drawing a blue marble.

Total marbles: $6 + 4 + 5 = 15$. $P(\text{blue}) = \dfrac{4}{15}$.

5 A fair six-sided die is rolled twice. Find the probability of rolling a 6 both times.

Since the rolls are independent: $P(6 \text{ and } 6) = \dfrac{1}{6} \times \dfrac{1}{6} = \dfrac{1}{36}$.

Combinations and permutations

6 In how many ways can a committee of 3 people be chosen from a group of 8, using combinations?

$\binom{8}{3} = \dfrac{8!}{3!\,5!} = \dfrac{8 \times 7 \times 6}{3 \times 2 \times 1} = 56$.

7 In how many ways can first, second, and third place be awarded among 6 runners, using permutations?

$P(6, 3) = \dfrac{6!}{3!} = 6 \times 5 \times 4 = 120$.

Conditional probability and two-way tables

8 A survey of 250 students found: 90 like math and science, 60 like only math, 50 like only science, 50 like neither. Find the probability a student likes science, given that they like math.

Total who like math: $90+60=150$. Of those, 90 also like science. $P(\text{science} \mid \text{math}) = \frac{90}{150} = 0.6$, or 60%.

Standard deviation and data spread

9 Two data sets have the same mean, but Data Set A has standard deviation 2 and Data Set B has standard deviation 9. Explain what this tells you about the two data sets.

A smaller standard deviation (Data Set A) means the values are clustered tightly around the mean, while a larger standard deviation (Data Set B) means the values are much more spread out — even though both sets share the same central tendency (mean).

Reading and interpreting data displays

10 A bar graph shows quarterly sales: Q1: 40, Q2: 55, Q3: 65, Q4: 50 (in thousands of dollars). Find the total annual sales and the percent of annual sales that occurred in Q3.

Total annual sales: $40+55+65+50=210$ thousand dollars. Percent from Q3: $\frac{65}{210} \times 100\% \approx 31\%$.

Expected value

11 A game costs 5 dollars to play. There is a 20% chance of winning 20 dollars and an 80% chance of winning nothing. Find the expected value of playing the game, and determine whether it is a fair game.

Expected winnings: $0.20(20)+0.80(0)=4$ dollars. Since the cost to play is 5 dollars but the expected winnings are only 4 dollars, the expected net value is $4-5=-1$ dollar — the game is not fair; on average, a player loses 1 dollar per play.

Multi-part probability and statistics problem

12 This question has two parts. A jar contains 5 red and 3 blue chips. Two chips are drawn without replacement. (a) Find the probability that both chips are red. (b) Find the probability that the two chips are different colors.

(a) $P(\text{first red}) = \frac{5}{8}$. $P(\text{second red} \mid \text{first red}) = \frac{4}{7}$. $P(\text{both red}) = \frac{5}{8} \times \frac{4}{7} = \frac{20}{56} = \frac{5}{14}$. (b) $P(\text{red then blue}) = \frac{5}{8} \times \frac{3}{7} = \frac{15}{56}$. $P(\text{blue then red}) = \frac{3}{8} \times \frac{5}{7} = \frac{15}{56}$. $P(\text{different colors}) = \frac{15}{56} + \frac{15}{56} = \frac{30}{56} = \frac{15}{28}$.

The complement rule and 'at least' probability

13 A fair coin is flipped 3 times. Find the probability of getting at least one heads, using the complement rule.

$P(\text{no heads}) = P(\text{all tails}) = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$. $P(\text{at least one heads}) = 1 - \frac{1}{8} = \frac{7}{8}$.

Quartiles and the five-number summary

14 A data set has the five-number summary: minimum 5, Q1 = 12, median 18, Q3 = 25, maximum 40. Find the interquartile range (IQR), and state what it measures.

IQR $= Q3 - Q1 = 25 - 12 = 13$. The IQR measures the spread of the middle 50% of the data, and is often used to identify outliers, since it is less sensitive to extreme values than the full range.

Independent events and the multiplication rule

15 A spinner has 4 equal sections (red, blue, green, yellow), and a separate fair die is rolled. Find the probability of spinning red AND rolling a number greater than 4.

$P(\text{red}) = \frac{1}{4}$. $P(\text{greater than 4}) = P(5 \text{ or } 6) = \frac{2}{6} = \frac{1}{3}$. Since independent:
$P(\text{red and} > 4) = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12}$.

A multi-part statistics problem with grouped data

16 This question has two parts. A teacher records test scores for two classes: Class A (25 students, mean 82) and Class B (15 students, mean 90). (a) Find the combined mean score for all 40 students. (b) If 5 more students join Class A with an average score of 70, find the new combined mean for all 45 students in the combined group.

(a) Total from A: $25(82) = 2050$. Total from B: $15(90) = 1350$. Combined total: $2050 + 1350 = 3400$. Combined mean: $\frac{3400}{40} = 85$. (b) New students' total: $5(70) = 350$. New grand total: $3400 + 350 = 3750$. New mean: $\frac{3750}{45} \approx 83.3$.

Probability using a tree diagram approach

17 A box has 4 white and 6 black balls. Two balls are drawn without replacement. Find the probability both balls are white.

$P(\text{first white}) = \frac{4}{10}$. $P(\text{second white} \mid \text{first white}) = \frac{3}{9}$. $P(\text{both white}) = \frac{4}{10} \times \frac{3}{9} = \frac{12}{90} = \frac{2}{15}$.